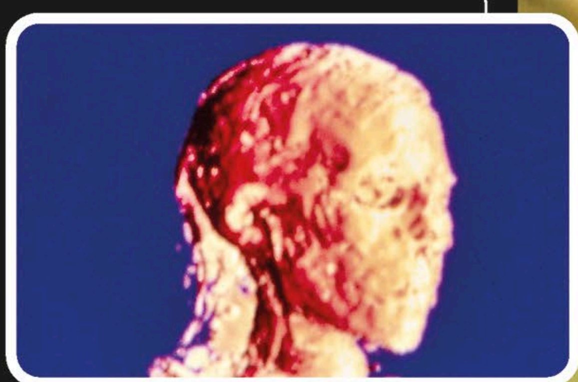
EXAMINING ARTIFACTS

CT SCANS

A CT (computed tomography) scanner can use X-rays to reveal the insides of an object, usually the organs of a human body. The machine sends X-rays through the object at different angles. Measurements of how much X-ray passes through are combined to produce a 3D picture. Doctors can use these machines to examine their patients, while archaeologists can see inside ancient artifacts such as mummies and sarcophagi without taking them apart.

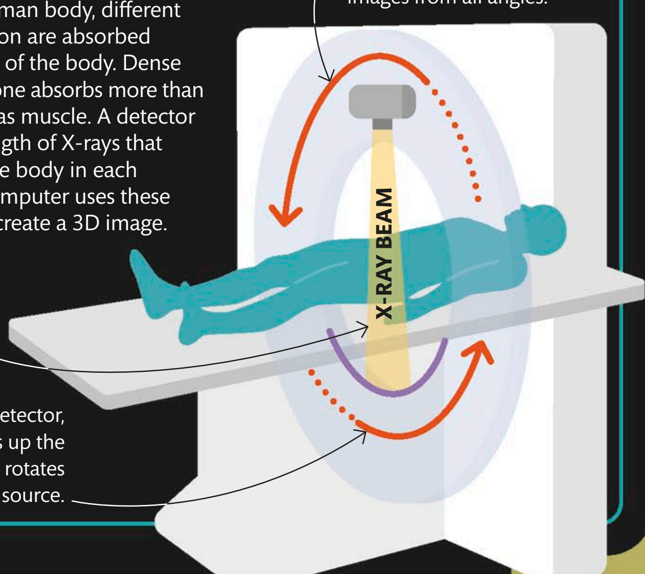
MAKING A MODEL

X-ray images are taken from all around the object and then processed by a computer to generate detailed cross-sectional images, called slices, and 3D models of structures within the object. Seen here is the computer-generated model of the head of an Ancient Egyptian mummy.



X-RAYS

X-rays are a type of invisible radiation (see page 142) that can pass through most materials. As X-rays travel through an object like a human body, different amounts of radiation are absorbed by different tissues of the body. Dense material such as bone absorbs more than soft material such as muscle. A detector measures the strength of X-rays that make it through the body in each direction, and a computer uses these measurements to create a 3D image.



Uncovering secrets

The 3,000-year-old mummy of Nesperennub, an Egyptian priest, is seen here being moved into a CT scanner. A study of the completed scans revealed a tiny pit on the inside of his skull, which suggests he might have died due to a major illness that affected his bones.